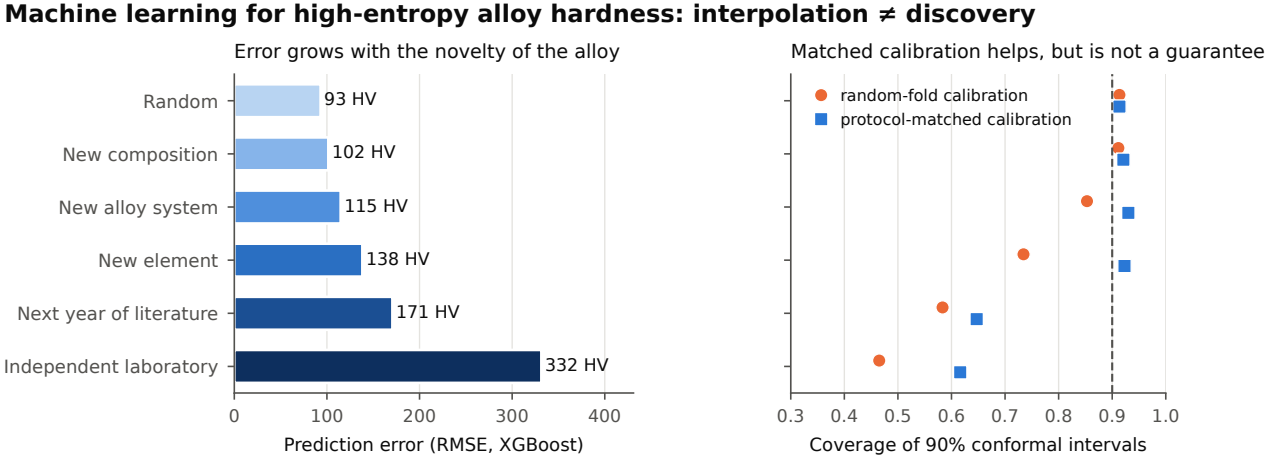
Machine learning for high-entropy alloy hardness: interpolation ≠ discovery
Error grows with the novelty of the alloy
Random
New composition
New alloy system
New element
Next year of literature
Independent laboratory
93 HV
102 HV
115 HV
138 HV
171 HV
332 HV
0
100
200
300
400
Prediction error (RMSE, XGBoost)
Matched calibration helps, but is not a guarantee
random-fold calibration
protocol-matched calibration
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Coverage of 90% conformal intervals